

Adaptive model predictive control for constrained nonlinear systems[☆]

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ABSTRACT

In this paper, a method is proposed for the adaptive model predictive control of constrained nonlinear system. Rather than relying on the inherent robustness properties of standard NMPC, the developed technique explicitly account for the transient effect of parametric estimation error by combining a parameter adjustment mechanism with robust MPC algorithms. The parameter estimation routine employed guarantees non-increase of the estimation error vector. This means that the controller employs a process model which approaches that of the true system over time. These estimates are used to update the parameter uncertainty set at every time step, resulting in a gradual reduction in the conservative and/or computational effects of the incorporated robust features.

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1. Introduction

Model predictive control (MPC) has attracted a great deal of interest from practitioners due to the relative ease constraints can be incorporated [1]. Although it has been proven that a standard implementation of MPC using a nominal model of the system dynamics exhibit nominal robustness to sufficiently small disturbances ([2,3]), such marginal robustness guarantee may be unacceptable in practical situations. Present uncertainties must be accounted for in the computation of the control law to achieve robust stability. One way to cope with uncertainty in the system model is to employ robust MPC methods, which explicitly account for system's uncertainties. Since robust controllers (in general) cannot learn changes in the plant, their performance is limited by the quality of the model plus the uncertainty description initially available. On the other hand, adaptive control has the potential to improve system performance as it updates the model online based on measurement data.

In general, most physical systems possess parametric uncertainties or unmeasurable parameters. Examples in chemical engineering include reaction rates, activation energies, fouling factors, and microbial growth rates. Since parametric uncertainty may degrade the performance of MPC, mechanisms to update the unknown or uncertain parameters are desirable in application. One

possibility would be to use state measurements to update the model parameters off-line. A more attractive possibility is to apply adaptive extensions of MPC in which parameter estimation and control are performed online.

While some results are available for linear adaptive MPC (See [4] for example), very few adaptive MPC schemes have been developed for nonlinear systems ([5,6]). The result in [6], implements a certainty equivalence nominal-model MPC feedback to stabilize a parametric uncertain system subject to an input constraint. The result shows that there must exist a finite time such that an excitation condition is satisfied and thus parameter convergence is achieved. There is no mechanism to decrease the identification period in any way and moreover, it is only by assumption that the true system trajectory remains bounded during the identification phase. In [5], an input-to-state stable control Lyapunov functions (iss-clf) is used to develop a MPC scheme that provides robust stabilization in the absence of parameter estimation algorithm, and ensures asymptotic stability of the closed loop with parameter adaptation. However, the work only deals with unconstrained nonlinear systems.

The design of adaptive nonlinear MPC schemes is very challenging because the “separation principle assumption” widely employed in linear control theory is not applicable to general class of nonlinear systems, in particular in the presence of constraints. A true adaptive nonlinear MPC algorithm must address the issue of robustness to model uncertainty while the estimator is evolving. Unfortunately, this may not be achieved without introducing extra degree of conservativeness and/or computational complexity in the controller calculations. The recent work [7] provided an adaptive robust MPC that deals with both state and input

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constraints within an adaptive framework. In the presented approach, a set valued description of the parametric uncertainty is directly adapted online to reduce the conservativeness of the solutions, especially with respect to the design of terminal penalty. The parameterization of the feedback MPC policy in terms of uncertainty set and the underlying min–max feedback MPC used in the study make the controller's computation very challenging. The result can be viewed as a conceptual result that focus on performance improvement rather than implementation.

In this paper, it is assumed that the uncertainties in the system are due to static nonlinearities, expressible in the form of unknown (constant) model parameters. The study addresses the problem of adaptive MPC and incorporates robust features to guarantee closed loop stability and constraint satisfaction. First, a min–max feedback nonlinear MPC scheme is combined with an adaptation mechanism for the uncertainty set. The formulation accounts for the effect of future parameter estimation and automatically injects some useful excitation into the closed loop system to aid in parameter identification. Second, the developed technique is extended to a less computational algorithm based on Lipschitz bounds. In both cases, the parameter estimation routine employed guarantees non-increase of the estimation error vector. Using this estimates to update the parameter uncertainty set at every time step, results in a gradual reduction in the conservativeness or computational demands of the algorithms.

The remainder of the paper is organized as follows. The problem description is given in Section 2 while the parameter identifier employed is outlined in Section 3. Our proposed robust adaptive MPC techniques are presented in Sections 4 and 5. Simulation results are shown in Section 6 and conclusions are given in Section 7.

Nomenclature and Definitions: $\lambda(M)$ denotes the smallest eigenvalue of matrix M . A continuous function $\mu : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is of class $\mathcal{K}_\infty$ if $\mu(0) = 0$, $\mu(\cdot)$ is strictly increasing on $\mathbb{R}^+$ and is radially unbounded.

2. Problem description

The system considered is the following nonlinear parameter affine system

$$\dot{x} = f(x, u) + g(x, u)\theta \triangleq \mathcal{F}(x, u, \theta) \quad (1)$$

$\theta \in \mathbb{R}^{n_\theta}$ is the unknown parameter vector whose entries may represent physically meaningful unknown model parameters or could be associated with any finite set of universal basis functions. It is assumed that θ is uniquely identifiable and lie within an initially known compact set $\Theta^0 \triangleq B(\theta^0, z_\theta^0)$, a ball described by an initial nominal estimate θ^0 and associated error bound $z_\theta^0 = \sup_{s \in \Theta^0} \|s - \theta^0\|$. The mapping $\mathcal{F} : \mathbb{R}^{n_x} \times \mathbb{R}^{n_u} \times \mathbb{R}^{n_\theta} \rightarrow \mathbb{R}^{n_x}$ is assumed to be locally Lipschitz with respect to its arguments and $\mathcal{F}(0, 0, \theta) = 0$. The state and the control input trajectories are assumed to be subject to pointwise constraints $x \in \mathbb{X} \in \mathbb{R}^{n_x}$ and $u \in \mathbb{U} \in \mathbb{R}^{n_u}$ respectively. The objective of the study is to (robustly) stabilize the plant by means of state feedback adaptive MPC. Optimality of the resulting trajectories are measured with respect to the accumulation of some stage cost $L(x, u) \geq 0$. The cost is assumed to be continuous, $L(0, 0) = 0$, and $L(x, u) \geq \mu_L(\|x, u\|)$, where μ_L is a $\mathcal{K}_\infty$ function.

3. Estimation of uncertainty

3.1. Parameter adaptation

In the following, we give a brief description of the identifier employed in the adaptive robust design framework. The procedure

assumes the state of the system $x(\cdot)$ is accessible for measurement but do not require the measurement or computation of the velocity state vector $\dot{x}(\cdot)$. The algorithm is independent of the control structure employed.

Let the state predictor for (1) be denoted as $\hat{x}$ and define

$$\dot{\hat{x}} = f(x, u) + g(x, u)\hat{\theta} + ke + w\hat{\theta}, \quad (2)$$

where $\hat{\theta}$ is the parameter estimate vector, $k > 0$ a design constant or $(n \times n)$ matrix, $e = x - \hat{x}$ the prediction error and w the output of the filter

$$\dot{w} = g(x, u) - kw, \quad w(t_0) = 0. \quad (3)$$

Denoting the parameter estimation error as $\tilde{\theta} = \theta - \hat{\theta}$, it follows from (1) and (2) that

$$\dot{e} = g(x, u)\tilde{\theta} - ke - w\tilde{\theta}. \quad (4)$$

Defining

$$\eta = e - w\tilde{\theta} \quad (5)$$

we have from (3) and (4) that η can be generated from

$$\dot{\eta} = -k\eta, \quad \eta(t_0) = e(t_0). \quad (6)$$

The parameter update law used in this work is given by

$$\dot{\hat{\theta}} = \gamma w^\top (e - \eta), \quad \hat{\theta}(t_0) = \theta^0 \quad (7)$$

with $\gamma = \gamma^\top > 0$.

Note that the adaptive update law (7) is driven by the parameter estimation error $\tilde{\theta}$ instead of tracking or prediction errors. Such adaptation law remains active until parameter convergence is achieved and may result in faster convergence than the traditional update laws that depends only on tracking or prediction error. The performance of the update mechanism is outlined in the following lemma.

Lemma 1. Let $Q \in \mathbb{R}^{n_\theta \times n_\theta}$ be generated from

$$\dot{Q} = w^\top w, \quad Q(t_0) = 0 \quad (8)$$

and define $\underline{\gamma} = \lambda(\gamma)$, a Lyapunov function $V_{\tilde{\theta}} = \frac{1}{2}\tilde{\theta}^\top \tilde{\theta}$, excitation index $\mathcal{E}(t) = \lambda(Q(t))$, and contraction factor

$$0 < \alpha(t) = \frac{1}{1 + \underline{\gamma}\mathcal{E}(t)} \leq 1, \quad (9)$$

the identifier (7) is such that the estimation error $\|\tilde{\theta}\| = \|\theta - \hat{\theta}\|$ is non-increasing and

$$V_{\tilde{\theta}}(t) \leq \alpha(t)V_{\tilde{\theta}}(t_0), \quad \text{for all } t > t_0. \quad (10)$$

Proof. The non-increase of $\|\tilde{\theta}\|$ follows from (5) and (7), which can be easily shown by noting that

$$\dot{V}_{\tilde{\theta}} \leq -\underline{\gamma}\tilde{\theta}^\top w^\top (e - \eta) = -\underline{\gamma}(e - \eta)^\top (e - \eta) \quad (11)$$

$$= -\underline{\gamma}\tilde{\theta}^\top w^\top w \tilde{\theta} \leq 0. \quad (12)$$

Moreover,

$$\begin{aligned} V_{\tilde{\theta}}(t) &= V_{\tilde{\theta}}(t_0) + \int_{t_0}^t \dot{V}_{\tilde{\theta}}(\tau) d\tau \\ &\leq V_{\tilde{\theta}}(t_0) - \underline{\gamma} \int_{t_0}^t \tilde{\theta}^\top(\tau) w(\tau)^\top w(\tau) \tilde{\theta}(\tau) d\tau \\ &\leq V_{\tilde{\theta}}(t_0) - \underline{\gamma} \lambda \left(\int_{t_0}^t w(\tau)^\top w(\tau) d\tau \right) \min_{\tau \in [t_0, t]} \|\tilde{\theta}(\tau)\|^2. \end{aligned}$$

Hence

$$V_{\tilde{\theta}}(t) \leq V_{\tilde{\theta}}(t_0) - \underline{\gamma}\mathcal{E}(t)V_{\tilde{\theta}}(t). \quad (13)$$

Rearranging (13) concludes the proof of part (1). ■

3.2. Set adaptation

The uncertainty set $\Theta \triangleq B(\hat{\theta}, z_\theta)$ is updated online by updating the parameter estimate $\hat{\theta}$ and its associated error bound $z_\theta = \sup_{s \in \Theta} \|s - \hat{\theta}\|$. The parameter adaptation law (7) gives $\hat{\theta}$ while z_θ is updated based on the observed parameter estimation error or system's excitation. The error bound is given as

$$z_\theta = \min(z_\theta^\varepsilon, z_\theta^{e\eta}) \quad (14)$$

with $z_\theta^{e\eta}$ and z_θ^ε obtained from

$$z_\theta^{e\eta} = \sqrt{V_{e\eta}}, \quad (15a)$$

$$\dot{V}_{e\eta} = -\underline{\gamma}(e - \eta)^T(e - \eta), \quad V_{e\eta}(t_0) = \frac{1}{2}(z_\theta^0)^2 \quad (15b)$$

and

$$z_\theta^\varepsilon = \sqrt{V_\varepsilon}, \quad (16a)$$

$$V_\varepsilon(t) = \alpha(t)V_\varepsilon(t_0), \quad V_\varepsilon(t_0) = \frac{1}{2}(z_\theta^0)^2 \quad (16b)$$

respectively.

The advantage of updating z_θ according to the system's excitation (16) over the observed error (15) is that contraction of z_θ can be triggered even when the actual parameter estimation error is zero.

Algorithm 1. The uncertainty set Θ is updated as follows:

- (1) Initialize $z_\theta(t_i) = z_\theta^0$, $\hat{\theta}(t_i) = \hat{\theta}^0$ and $\Theta(t_i) = B(\hat{\theta}(t_i), z_\theta(t_i))$.
- (2) When the following condition is (first) satisfied at an arbitrary time t_{i+1} ,

$$z_\theta(t_{i+1}) \leq z_\theta(t_i) - \|\hat{\theta}(t_{i+1}) - \hat{\theta}(t_i)\|.$$

Perform the update

$$\Theta(t_{i+1}) = B(\hat{\theta}(t_{i+1}), z_\theta(t_{i+1})).$$

- (3) Iterate back to step 2, incrementing $i = i + 1$.

It will be shown in the following lemma that the uncertainty set Θ when implemented according to Algorithm 1 contracts in a strictly nested fashion without excluding θ .

Lemma 2. The evolution of $\Theta \triangleq B(\hat{\theta}, z_\theta)$ under (7), (14)–(16) and Algorithm 1 is such that

- (i) $\Theta(t_2) \subseteq \Theta(t_1)$, $t_0 \leq t_1 \leq t_2$
- (ii) $\theta \in \Theta(t_0) \Rightarrow \theta \in \Theta(t)$, $\forall t \geq t_0$.

Proof. (i) If $\Theta(t_{i+1}) \not\subseteq \Theta(t_i)$, then

$$\sup_{s \in \Theta(t_{i+1})} \|s - \hat{\theta}(t_i)\| \geq z_\theta(t_i). \quad (17)$$

However, it follows from triangle inequality and Algorithm 1 that Θ , at update times, obeys

$$\begin{aligned} & \sup_{s \in \Theta(t_{i+1})} \|s - \hat{\theta}(t_i)\| \\ & \leq \sup_{s \in \Theta(t_{i+1})} \|s - \hat{\theta}(t_{i+1})\| + \|\hat{\theta}(t_{i+1}) - \hat{\theta}(t_i)\| \\ & \leq z_\theta(t_{i+1}) + \|\hat{\theta}(t_{i+1}) - \hat{\theta}(t_i)\| \leq z_\theta(t_i), \end{aligned}$$

which contradicts (17). Hence, Θ update guarantees $\Theta(t_{i+1}) \subseteq \Theta(t_i)$ and the strict contraction claim follows from the fact that Θ is held constant over update intervals $\tau \in (t_i, t_{i+1})$.

- (ii) The θ inclusion claim is proven by showing that

$$\|\tilde{\theta}(t)\| \leq z_\theta(t) \leq z_\theta(t_0) \quad \forall t \geq t_0 \quad (18)$$

which in turn establish that $\theta \in \Theta(t_0) \Rightarrow \theta \in B(\hat{\theta}(t), z_\theta(t))$, $\forall t \geq t_0$. We know that $V_{\hat{\theta}}(t_0) \leq V_{e\eta}(t_0) =$

$V_\varepsilon(t_0)$ (by definition) and it follows from (11) and (15) that $V_{\hat{\theta}}(t) \leq V_{e\eta}(t)$. Hence, by comparison lemma and the monotonous decrease of $V_{\hat{\theta}}$, we have

$$V_{\hat{\theta}}(t) \leq V_{e\eta}(t) \leq V_{e\eta}(t_0), \quad \forall t \geq t_0. \quad (19)$$

Similarly, it follows from (10) and (16) and the fact that $\alpha(t) \leq 1$ that

$$V_{\hat{\theta}}(t) \leq V_\varepsilon(t) \leq V_\varepsilon(t_0), \quad \forall t \geq t_0. \quad \blacksquare \quad (20)$$

4. Robust adaptive MPC—A min–max approach

In this section, the concept of min–max robust MPC is employed to provide robustness for the MPC controller during the adaptation phase. The resulting optimization problem can either be solved in open-loop or closed-loop. In the presented approach, we choose the least conservative option by performing optimization with respect to closed loop strategies. As in typical feedback-MPC fashion, the controller chooses input u as a function of the current states. The formulation consists of maximizing a cost function with respect to θ and minimizing over feedback control policies κ .

The receding horizon control law is defined by

$$u = \kappa_{mpc}(x, \hat{\theta}, z_\theta) \triangleq \kappa^*(0, x, \hat{\theta}, z_\theta) \quad (21a)$$

$$\kappa^* \triangleq \arg \min_{\kappa(\cdot, \cdot, \cdot, \cdot)} J(x, \hat{\theta}, z_\theta, \kappa) \quad (21b)$$

where $J(x, \hat{\theta}, z_\theta, \kappa)$ is the (worst-case) cost associated with the optimal control problem:

$$J(x, \hat{\theta}, z_\theta, \kappa) \triangleq \max_{\theta \in \Theta} \int_0^T L(x^p, u^p) d\tau + W(x^p(T), \tilde{\theta}^p(T)) \quad (22a)$$

s.t. $\forall \tau \in [0, T]$

$$\dot{x}^p = f(x^p, u^p) + g(x^p, u^p)\theta, \quad x^p(0) = x \quad (22b)$$

$$\dot{w}^p = g^T(x^p, u^p) - kw^p, \quad w^p(0) = w \quad (22c)$$

$$\dot{\hat{\theta}}^p = \gamma w^p T w^p (\theta - \hat{\theta}^p), \quad \hat{\theta}^p(0) = \hat{\theta} \quad (22d)$$

$$u^p(\tau) \triangleq \kappa(\tau, x^p(\tau), \hat{\theta}^p(\tau)) \in \mathbb{U} \quad (22e)$$

$$x^p(\tau) \in \mathbb{X}, \quad x^p(T) \in \mathbb{X}_f(\tilde{\theta}^p(T)). \quad (22f)$$

In the formulation, the effect of future parameter adaptation is accounted for, which results in less conservative worst-case predictions. Also, the conservativeness of the terminal cost is reduced by parameterizing both W and $\mathbb{X}_f$ as functions of $\tilde{\theta}(T)$. Parameterizing the terminal penalty as a function of $\tilde{\theta}$ ensures that the algorithm will seek to reduce the parameter error in the process of optimizing the cost function J .

4.1. Implementation algorithm

Algorithm 2. The MPC algorithm performs as follows: At sampling instant t_i

- (1) Measure the current state of the plant $x(t)$ and obtain the current value of matrix w from (3)–(16).
- (2) Update the parameter estimates $\hat{\theta}$ and uncertainty bound z_θ via (7) and (14) as follows:

$$\begin{aligned} & (\hat{\theta}, z_\theta) \\ & = \begin{cases} (\hat{\theta}(t_i), z_\theta(t_i)), & \text{if } z_\theta(t_i) \leq z_\theta(t_{i-1}) - \|\hat{\theta}(t_i) - \hat{\theta}(t_{i-1})\| \\ (\hat{\theta}(t_{i-1}), z_\theta(t_{i-1})), & \text{otherwise.} \end{cases} \end{aligned}$$

- (3) Solve the optimization problem (21) and (22) and apply the resulting feedback control law to the plant until the next sampling instant.

- (4) Repeat the procedure from step 1 for the next sampling instant, incrementing $i = i + 1$.

In the remainder of this section, we drop constraints (22e) and (22f) by using the convention that if some of the constraints are not satisfied, then the value of J is $+\infty$.

4.2. Closed loop robust stability

Robust stability is guaranteed if predicted state at terminal time belong to a robustly invariant set for all possible uncertainties. Let $\tilde{\Theta}^0 = \{\tilde{\theta} \mid \|\tilde{\theta}\| \leq z_\theta(t_0)\}$, sufficient conditions for the robust MPC (22) to guarantee stabilization of the origin is outlined below:

Criterion 3. The terminal penalty function $W : \mathbb{X}_f \times \tilde{\Theta}^0 \rightarrow [0, +\infty]$ and the terminal constraint function $\mathbb{X}_f : \tilde{\Theta}^0 \rightarrow \mathbb{X}$ are such that for each $(\theta, \hat{\theta}, \tilde{\theta}) \in (\Theta^0 \times \Theta^0 \times \tilde{\Theta}^0)$, there exists a feedback $k_f(\cdot, \hat{\theta}) : \mathbb{X}_f \rightarrow \mathbb{U}$ satisfying

- (1) $0 \in \mathbb{X}_f(\tilde{\theta}) \subseteq \mathbb{X}$, $\mathbb{X}_f(\tilde{\theta})$ closed
- (2) $k_f(x, \hat{\theta}) \in \mathbb{U}$, $\forall x \in \mathbb{X}_f(\tilde{\theta})$
- (3) $W(x, \tilde{\theta})$ is continuous with respect to $x \in \mathbb{R}^n$
- (4) $\mathbb{X}_f(\tilde{\theta})$ is strongly positively invariant under $u_L \triangleq k_f(x, \hat{\theta})$ with respect to the differential inclusion $\dot{x} \in \mathcal{F} \triangleq f(x, u_L) + g(x, u_L)\Theta$
- (5) $W(x(t+\delta), \tilde{\theta}(t)) - W(x(t), \tilde{\theta}(t)) \leq -\int_t^{t+\delta} L(x, u_L)d\tau$, $\forall x \in \mathbb{X}_f(\tilde{\theta})$.

In addition to Criterion 3, the $\tilde{\theta}$ dependence of W and $\mathbb{X}_f$ is required to satisfy the following:

Criterion 4. For any $\tilde{\theta}_1, \tilde{\theta}_2 \in \tilde{\Theta}^0$ s.t. $\|\tilde{\theta}_2\| \leq \|\tilde{\theta}_1\|$,

- (1) $W(x, \tilde{\theta}_2) \leq W(x, \tilde{\theta}_1)$, $\forall x \in \mathbb{X}_f(\tilde{\theta}_1)$
- (2) $\mathbb{X}_f(\tilde{\theta}_2) \supseteq \mathbb{X}_f(\tilde{\theta}_1)$.

Criterion 3 is an extension of the general sufficient condition for closed-loop stability of nonlinear MPC [8,9]. Note that the criterion require only the existence, not knowledge, of $k_f(x, \hat{\theta})$ and the stability condition requires the terminal penalty function $W(x, \tilde{\theta})$ to be a robust-CLF on the domain $\mathbb{X}_f(\tilde{\theta})$. Since the controller is only required to stabilize the system in a neighbourhood of the steady state, a linear approximation of the nonlinear system can be used to construct the design parameters. There are well established methods for computing robust controllers for linear systems. Its associated quadratic Lyapunov function can be chosen as terminal cost and a level set can be used as terminal set.

Criterion 4 is specific for the proposed approach. The condition can be satisfied by a CLF that involves a quadratic function of $\tilde{\theta}$. When a CLF that is independent of $\tilde{\theta}$ is used, the criteria will be satisfied with equality.

Theorem 5. Let $X_0 \triangleq X_0(\Theta^0) \subseteq \mathbb{X}$ denote the set of initial states with uncertainty Θ^0 for which (21) has a solution. Assuming Criteria 3 and 4 are satisfied, then the closed loop system state x , given by (1)–(3), (6), (7), (14)–(16) and (21), originating from any $x_0 \in X_0$ feasibly approaches the origin as $t \rightarrow +\infty$.

The proof can be found in the Appendix.

5. Robust adaptive MPC—A Lipschitz based approach

Due to the computational complexity associated with (feedback) min–max optimization problem for non-linear systems, it is (sometimes) more practical to use a more conservative but computationally efficient methods. Examples of such approaches include

Lipschitz based methods [2,10] and those based on the concept of reachable sets [11].

In this section, we present a Lipschitz based method whereby the nominal model rather than the unknown bounded system state is controlled, subject to conditions that ensure that given constraints are satisfied for all possible uncertainties. State prediction error bound is determined based on the Lipschitz continuity of the model. A knowledge of appropriate Lipschitz bounds for the x -dependence of the dynamics $f(x, u)$ and $g(x, u)$ are assumed as follows:

Assumption 6. A set of functions $\mathcal{L}_j : \mathbb{X} \times \mathbb{U} \rightarrow \mathbb{R}^+$, $j \in \{f, g\}$ are known which satisfy

$$\mathcal{L}_j(\mathbb{X}, u) \geq \min \left\{ \mathcal{L}_j \left| \sup_{x_1, x_2 \in \mathbb{X}} (\|j(x_1, u) - j(x_2, u)\| - \mathcal{L}_j \|x_1 - x_2\|) \leq 0 \right. \right\},$$

where for $j \equiv g$ is interpreted as an induced norm.

5.1. Prediction of state error bound

In order to consider the effect of the uncertainty $\tilde{\theta}$ in the controller synthesis, we have to compute a bound on the difference between the nominal state trajectory and the solution of the actual system

$$\dot{x} = f(x, u) + g(x, u)\theta, \quad (23)$$

and the nominal model controlled by the same input u

$$\dot{x}^p = f(x^p, u) + g(x^p, u)\hat{\theta}, \quad (24)$$

it follows that

$$\begin{aligned} \|\dot{x} - \dot{x}^p\| &= \|f(x, u) - f(x^p, u)\| + \|g(x, u)\theta - g(x^p, u)\theta\| \\ &\quad + \|g(x^p, u)\theta - g(x^p, u)\hat{\theta}\| \\ &= \mathcal{L}_f \|x - x^p\| + \mathcal{L}_g \|\theta\| \|x - x^p\| + \|g(x^p, u)\| \|\theta - \hat{\theta}\|. \end{aligned}$$

Therefore, a worst-case deviation $z_x^p \geq \max_{\theta \in \Theta} \|x - x^p\|$ can be generated from

$$\dot{z}_x^p = (\mathcal{L}_f + \mathcal{L}_g \Pi) z_x^p + \|g(x^p, u)\| z_\theta, \quad z_x^p(t_0) = 0 \quad (25)$$

where $\Pi = z_\theta + \|\hat{\theta}\|$.

5.2. Lipschitz based finite horizon optimal control problem

The model predictive feedback is defined as

$$u = \kappa_{mpc}(x, \hat{\theta}, z_\theta) = u^*(0) \quad (26a)$$

$$u^*(\cdot) \triangleq \arg \min_{u_{[0,T]}^p} J(x, \hat{\theta}, z_\theta, u^p) \quad (26b)$$

where $J(x, \hat{\theta}, z_\theta, u^p)$ is given by the optimal control problem:

$$J(x, \hat{\theta}, z_\theta, u^p) = \int_0^T L(x^p, u^p) d\tau + W(x^p(T), z_\theta^p) \quad (27a)$$

s.t. $\forall \tau \in [0, T]$

$$\dot{x}^p = f(x^p, u^p) + g(x^p, u^p)\hat{\theta}, \quad x^p(0) = x \quad (27b)$$

$$\dot{z}_x^p = (\mathcal{L}_f + \mathcal{L}_g \Pi) z_x^p + \|g(x^p, u^p)\| z_\theta, \quad z_x^p(0) = 0 \quad (27c)$$

$$X^p(\tau) \triangleq B(x^p(\tau), z_x^p(\tau)) \subseteq \mathbb{X}, \quad u^p(\tau) \in \mathbb{U} \quad (27d)$$

$$X^p(T) \subseteq \mathbb{X}_f(z_\theta^p). \quad (27e)$$

In the proposed formulation, the uncertainty radius z_θ which appears in (27c) and in the terminal expressions of (27a) and (27e) are held constant over the prediction horizon. However, they are updated at every sampling instant. This enlarges the terminal domain and hence reduces the conservativeness of the robust MPC that would otherwise have been designed based on a large initial uncertainty $z_\theta(t_0)$.

5.3. Implementation algorithm

Algorithm 3. The Lipschitz based MPC algorithm performs as follows: At sampling instant t_i

- (1) Measure the current state of the plant $x(t)$.
- (2) Update the parameter estimates $\hat{\theta}$ and uncertainty bound z_θ via (7) and (14) as follows:

$$\begin{aligned} & (\hat{\theta}, z_\theta) \\ &= \begin{cases} (\hat{\theta}(t_i), z_\theta(t_i)), & \text{if } z_\theta(t_i) \leq z_\theta(t_{i-1}) - \|\hat{\theta}(t_i) - \hat{\theta}(t_{i-1})\| \\ (\hat{\theta}(t_{i-1}), z_\theta(t_{i-1})), & \text{otherwise.} \end{cases} \end{aligned}$$

- (3) Solve the optimization problem (26) and (27) and apply the resulting feedback control law to the plant until the next sampling instant.
- (4) Repeat the procedure from step 1 for the next sampling instant, incrementing $i = i + 1$.

Theorem 7. Let $X'_0 \triangleq X'_0(\theta^0) \subseteq \mathbb{X}$ denote the set of initial states for which (26) has a solution. Assuming that Assumption 6 and Criteria 3 and 4 are satisfied, then the origin of the closed loop system given by (1)–(3), (6), (14)–(16) and (26) is feasibly asymptotically stabilized from any $x_0 \in X'_0$.

The proof of the theorem is given in the Appendix.

6. Simulation example

Consider the regulation of a continuous stirred tank reactor where a first order, irreversible exothermic reaction $\mathbf{A} \rightarrow \mathbf{B}$ is carried out. Assuming constant liquid level, the reaction is described by the following dynamic model [12]:

$$\begin{aligned} \dot{C}_A &= \frac{q}{V} (C_{Ain} - C_A) - k_0 \exp\left(\frac{-E}{RT_r}\right) C_A \\ \dot{T}_r &= \frac{q}{V} (T_{in} - T_r) - \frac{\Delta H}{\rho c_p} k_0 \exp\left(\frac{-E}{RT_r}\right) C_A \\ &\quad + \frac{UA}{\rho c_p V} (T_c - T_r). \end{aligned}$$

The states C_A and T_r are the concentrations of components A and the reactor temperature respectively. The manipulated variable T_c is temperature of the coolant stream.

It is assumed that reaction kinetic constant k_0 and heat of reaction ΔH are only nominally known and parameterized as $k_0 = \theta_1 \times 10^{10} \text{ min}^{-1}$ and $\Delta H k_0 = -\theta_2 \times 10^{15} \text{ J/mol min}$ with the parameters satisfying $0.1 \leq \theta_1 \leq 10$ and $0.1 \leq \theta_2 \leq 10$. The objective is to adaptively regulate the unstable equilibrium $C_A^{eq} = 0.5 \text{ mol/l}$, $T_r^{eq} = 350 \text{ K}$, $T_c^{eq} = 300 \text{ K}$ while satisfying the constraints $0 \leq C_A \leq 1$, $280 \leq T_r \leq 370$ and $280 \leq T_c \leq 370$. The nominal operating conditions, which corresponds to the given unstable equilibrium are taken from [12]: $q = 100 \text{ l/min}$, $V = 100 \text{ l}$, $\rho = 1000 \text{ g/l}$, $c_p = 0.239 \text{ J/g K}$, $E/R = 8750 \text{ K}$, $UA = 5 \times 10^4 \text{ J/min K}$, $C_{Ain} = 1 \text{ mol/l}$ and $T_{in} = 350 \text{ K}$.

Defining $x = [\frac{C_A - C_A^{eq}}{0.5}, \frac{T_r - T_r^{eq}}{20}]'$, $u = \frac{T_c - T_c^{eq}}{20}$, the stage cost $L(x, u)$ was selected as a quadratic function of its arguments:

$$L(x, u) = x^T Q_x x + u^T R_u u \quad (28a)$$

$$Q_x = \begin{bmatrix} 0.5 & 0 \\ 0 & 1.1429 \end{bmatrix} \quad R_u = 1.333. \quad (28b)$$

The terminal penalty function used is a quadratic parameter-dependent Lyapunov function $W(x, \theta) = x^T P(\theta) x$ for the linearized system. Denoting the closed-loop system under a local robust stabilizing controller $u = k_f(\theta) x$ as $\dot{x} = A_{cl}(\theta) x$. The matrix $P(\theta) := P_0 + \theta_1 P_1 + \theta_2 P_2 + \dots + \theta_{n\theta} P_{n\theta}$ was selected to satisfy the Lyapunov system of LMIs

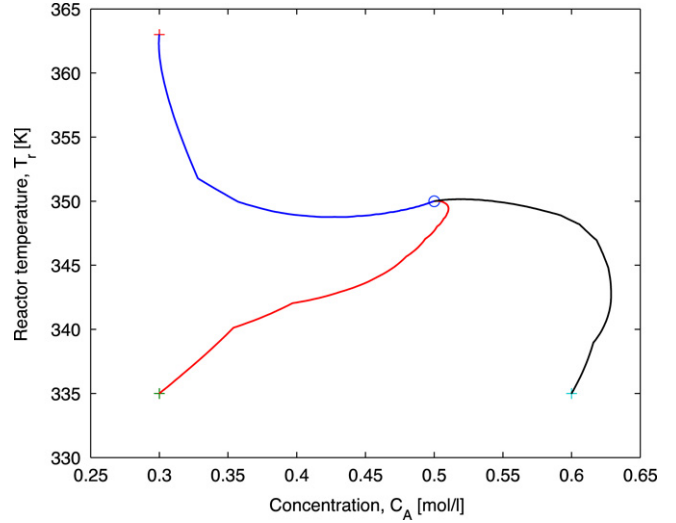


Fig. 1. Closed-loop state trajectories.

$$P(\theta) > 0$$

$$A_{cl}(\theta)^T P(\theta) + P(\theta) A_{cl}(\theta) < 0$$

for all admissible values of θ . Since θ lie between known extrema values, the task of finding $P(\theta)$ reduces to solving a finite set of linear matrix inequalities by introducing additional constraints [13]. For the initial nominal estimate $\theta^0 = 5.05$ and $z_\theta^0 = 4.95$, the matrix $P(\theta^0)$ obtained is

$$P(\theta^0) = \begin{bmatrix} 0.6089 & 0.1134 \\ 0.1134 & 4.9122 \end{bmatrix} \quad (29)$$

and the corresponding terminal region is

$$\mathbb{X}_f = \{x : x^T P(\theta^0) x \leq 0.25\}. \quad (30)$$

For simulation purposes, the true values of the unknown parameters were chosen as $k_0 = 7.2 \times 10^{10} \text{ min}^{-1}$ and $\Delta H = -5.0 \times 10^4 \text{ J/mol}$ which implies $\theta_1 = 7.2$ and $\theta_2 = 3.6$. The Lipschitz-based approach was used for the controller calculations and the result was implemented according to Algorithm 3. Since the regressor matrix for this reactor model is diagonal, we define uncertainty bound z_θ for each parameter estimate and adapt the pairs $(\hat{\theta}_1, z_{\theta_1})$ and $(\hat{\theta}_2, z_{\theta_2})$ separately.

The system was simulated from three different initial states $(C_A(0), T_r(0)) = (0.3, 335)$, $(0.6, 335)$ and $(0.3, 363)$. The closed-loop trajectories are reported in Figs. 1–4. The results demonstrate that the adaptive MPC regulates the system states to the open loop unstable equilibrium values and satisfies the imposed state and input constraints. The parameter estimates converge to the true values and the uncertainty bound z_θ reduces over time.

7. Conclusions

In this chapter, we presented an adaptive MPC design technique for constrained nonlinear systems with parametric uncertainties. The system's performance is improved over time as the adaptive control updates the model online. The controller parameters is updated only when an improved parameter estimate is obtained. Robustly stabilizing MPC schemes are incorporated to ensure robustness of the algorithm to parameter estimation error during the adaptation phase. The two robust approaches, min-max and Lipschitz-based method, presented provides a tradeoff between computational complexity and conservatism of the solutions. In both cases, the controller is designed in such a way that the computational requirement/conservativeness of the robust adaptive MPC reduces with reduction in parameter uncertainty. Extension of the approach to systems with both parametric uncertainties and time varying disturbances is ongoing.

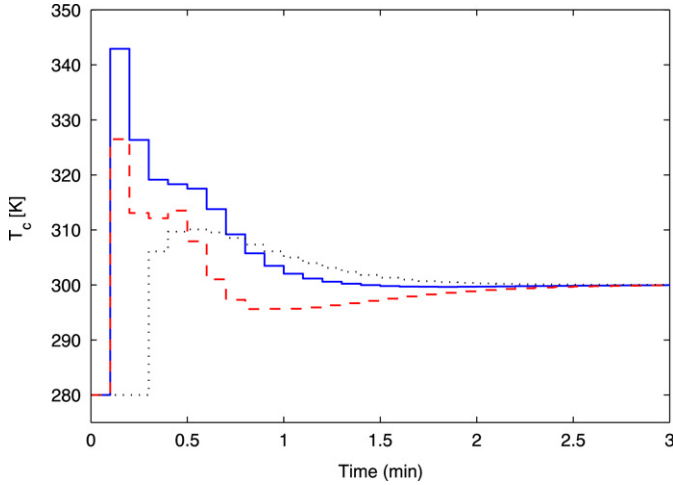


Fig. 2. Closed-loop input profiles for states starting at different initial conditions ($C_A(0)$, $T_r(0)$): (0.3, 335) is solid line, (0.6, 335) is dashed line and (0.3, 363) is the dotted line.

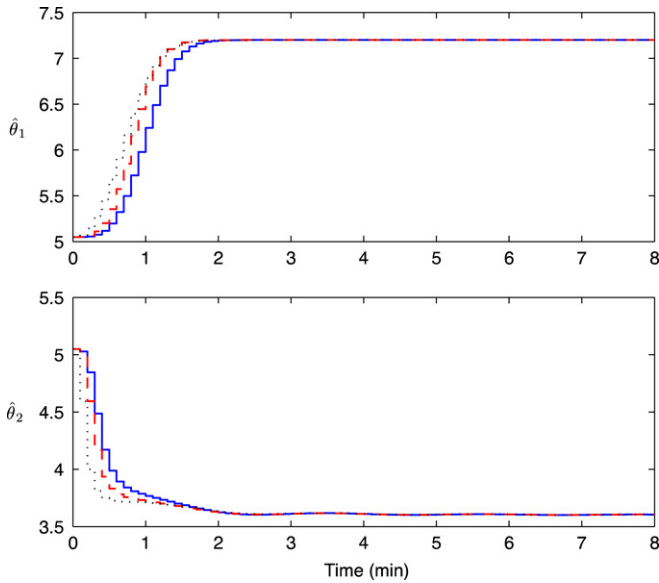


Fig. 3. Closed-loop parameter estimates profile for states starting at different initial conditions ($C_A(0)$, $T_r(0)$): (0.3, 335) is solid line, (0.6, 335) is dashed line and (0.3, 363) is the dotted line.

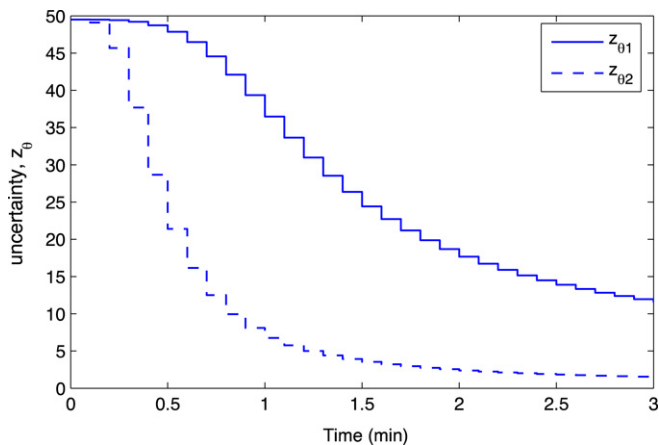


Fig. 4. Closed-loop uncertainty bound trajectories for initial condition (C_A , T_r) = (0.3, 335).

Appendix

A.1. Proof of Theorem 5

Feasibility: The closed-loop stability is based upon the feasibility of the control action at each sample time. Assuming, at time t , that an optimal solution $u_{[0,T]}^p$ to the optimization problem (21) exist and is found. Let Θ^p denote the estimated uncertainty set at time t and Θ^v denote the set at time $t + \delta$ that would result with the feedback implementation of $u_{[t,t+\delta]} = u_{[0,\delta]}^p$. Also, let x^p represents the worst case state trajectory originating from $x^p(0) = x(t)$ and x^v represents the trajectory originating from $x^v(0) = x + \delta v$ under the same feasible control input $u_{[\delta,T]}^v = u_{[\delta,T]}^p$. Moreover, let $X_{\Theta^b}^a \triangleq \{x^a | \dot{x}^a \in \mathcal{F}(x^a, u^p, \Theta^b) \triangleq f(x^a, u^p) + g(x^a, u^p)\Theta^b\}$.

Since the $u_{[0,T]}^p$ is optimal with respect to the worst case uncertainty scenario, it suffice to say that $u_{[0,T]}^p$ drives any trajectory $x^p \in X_{\Theta^p}^p$ into the terminal region $\mathbb{X}_f^p$. Since Θ is non-expanding over time, we have $\Theta^v \subseteq \Theta^p$ implying $x^v \in X_{\Theta^v}^p \subseteq X_{\Theta^p}^p$. The terminal region $\mathbb{X}_f^p$ is strongly positively invariant for the nonlinear system (1) under the feedback $k_f(\cdot, \cdot)$, the input constraint is satisfied in $\mathbb{X}_f^p$ and $\mathbb{X}_f^v \supseteq \mathbb{X}_f^p$ by criteria 2.2, 2.4 and 3.2 respectively. Hence, the input $u = [u_{[\delta,T]}^p, k_{f[T,T+\delta]}]$ is a feasible solution of (21) at time $t + \delta$ and by induction, the optimization problem is feasible for all $t \geq 0$.

Stability: The stability of the closed-loop system is established by proving strict decrease of the optimal cost $J^*(x, \hat{\theta}, z_\theta) \triangleq J(x, \hat{\theta}, z_\theta, \kappa^*)$. Let the trajectories $(x^p, \hat{\theta}^p, \tilde{\theta}^p, z_\theta^p)$ and control u^p correspond to any worst case minimizing solution of $J^*(x, \hat{\theta}, z_\theta)$. If $x_{[0,T]}^p$ were extended to $\tau \in [0, T + \delta]$ by implementing the feedback $u(\tau) = k_f(x^p(\tau), \hat{\theta}^p(\tau))$ on $\tau \in [T, T + \delta]$, then Criterion 3(5) guarantees the inequality

$$\int_T^{T+\delta} L(x^p, k_f(x^p, \hat{\theta}^p)) d\tau + W(x_{T+\delta}^p, \tilde{\theta}_T^p) - W(x_T^p, \tilde{\theta}_T^p) \leq 0 \quad (31)$$

where in (31) and in the remainder of the proof, $x_\sigma^p \triangleq x^p(\sigma)$, $\tilde{\theta}_\sigma^p \triangleq \tilde{\theta}^p(\sigma)$, for $\sigma = T, T + \delta$.

The optimal cost $J^*(x, \hat{\theta}, z_\theta)$

$$\begin{aligned} &= \int_0^T L(x^p, u^p) d\tau + W(x_T^p, \tilde{\theta}_T^p) \\ &\geq \int_0^T L(x^p, u^p) d\tau + W(x_T^p, \tilde{\theta}_T^p) \\ &\quad + \int_T^{T+\delta} L(x^p, k_f(x^p, \hat{\theta}^p)) d\tau + W(x_{T+\delta}^p, \tilde{\theta}_T^p) - W(x_T^p, \tilde{\theta}_T^p) \quad (32) \end{aligned}$$

$$\begin{aligned} &\geq \int_0^\delta L(x^p, u^p) d\tau + \int_\delta^T L(x^p, u^p) d\tau \\ &\quad + \int_T^{T+\delta} L(x^p, k_f(x^p, \hat{\theta}^p)) d\tau + W(x_{T+\delta}^p, \tilde{\theta}_{T+\delta}^p) \quad (33) \end{aligned}$$

$$\geq \int_0^\delta L(x^p, u^p) d\tau + J^*(x(\delta), \hat{\theta}(\delta), z_\theta(\delta)). \quad (34)$$

Then, it follows from (34) that

$$\begin{aligned} J^*(x(\delta), \hat{\theta}(\delta), z_\theta(\delta)) - J^*(x, \hat{\theta}, z_\theta) &\leq - \int_0^\delta L(x^p, u^p) d\tau \\ &\leq - \int_0^\delta \mu_L(\|x, u\|) d\tau \quad (35) \end{aligned}$$

where μ_L is a class $\mathcal{K}_\infty$ function. Hence $x(t) \rightarrow 0$ asymptotically.

Remark 8. In the above proof,

- (32) is obtained using inequality (31)
- (33) follows from Criterion 4.1 and the fact that $\|\tilde{\theta}\|$ is non-increasing
- (34) follows by noting that the last 3 terms in (33) is a (potentially) suboptimal cost on the interval $[\delta, T + \delta]$ starting from the point $(x^p(\delta), \hat{\theta}^p(\delta))$ with associated uncertainty set $B(\hat{\theta}^p(\delta), z_\theta^p(\delta))$.

A.2. Proof of Theorem 7

Feasibility: Let $u_{[0,T]}^p$ denotes the initial optimal or feasible solution of (26) and let $X^p \triangleq B(x^p, z_x^p)$ denotes the corresponding predicted ball of possible trajectories that starts at $(x^p, z_x^p, z_\theta^p)|_{\tau=0} = (x(t), 0, z_\theta(t))$. Similarly, let $X^v \triangleq B(x^v, z_x^v)$ denote the resulting cone originating from $(x^v, z_x^v, z_\theta^v)|_{\tau=\delta} = (x + \delta v, 0, z_\theta(t + \delta))$, under the same feasible control input $u_{[\delta,T]}^v = u_{[\delta,T]}^p$.

At time $\tau = 0$, it follows from (27b) and (27c) that $\dot{x}^p$ provides an upper bound on $\|\dot{x}^p - \dot{x}\|$, for $\dot{x} \in \mathcal{F}(x, u^p, \Theta)$. Thus the continuity of the trajectories $x^p(\tau)$ and $z_x^p(\tau)$ ensures that for a small enough $\delta > 0$, $x + \delta v \in X^p(\delta)$. Therefore, we assume w.l.o.g. that $x^v(\delta) \in X^p(\delta)$. Since $z_x^v(\delta) = 0$, it follows that $X^v(\delta) \subseteq X^p(\delta)$ and $\|x^p(\delta) - x^v(\delta)\| \leq z_x^p(\delta) - z_x^v(\delta)$.

Next, we establish that the inclusion $X^v(\tau) \subseteq X^p(\tau)$ holds for all $\tau \in [\delta, T]$ by showing that $\|x^p(\tau) - x^v(\tau)\| \leq z_x^p(\tau) - z_x^v(\tau)$, $\forall \tau \in [\delta, T]$. Defining the variable $e_z \triangleq z_x^p - z_x^v - \|x^p - x^v\|$, the dynamics of e_z satisfy

$$\begin{aligned} \dot{e}_z &= \dot{z}_x^p - \dot{z}_x^v - \|\dot{x}^p - \dot{x}^v\| \\ &= (\mathcal{L}_f + \mathcal{L}_g \Pi) z_x^p - (\mathcal{L}_f + \mathcal{L}_g \Pi) z_x^v \\ &\quad - \|f(x^p, u) - f(x^v, u) + (g(x^p, u) - g(x^v, u))\hat{\theta}\| \\ &\geq (\mathcal{L}_f + \mathcal{L}_g \Pi) (z_x^p - z_x^v) - \left(\mathcal{L}_f + \mathcal{L}_g \|\hat{\theta}\| \right) \|x^p - x^v\| \\ &\geq (\mathcal{L}_f + \mathcal{L}_g \Pi) e_z, \end{aligned}$$

from which the initial condition $e_z(\delta) \geq 0$ guarantees that $e_z(\tau) \geq 0$, $\forall \tau \in [\delta, T]$. This implies that $X^v(\tau) \subseteq X^p(\tau) \subseteq \mathbb{X}$ and $X^v(T) \subseteq$

$X^p(T) \subseteq \mathbb{X}_f^p$. Moreover, from the non-increase of the uncertainty bound z_θ , we have $z_\theta^v \leq z_\theta^p$ which implies that $\mathbb{X}_f^p \subseteq \mathbb{X}_f^v$. Therefore, the input $u = [u_{[\delta,T]}^p, k_{f[T,T+\delta]}]$ serves as a feasible solution for (26) at time $t + \delta$ and the feasibility result can be achieved by induction.

Stability: Closed loop stability is established by showing that the optimal value function is non-increasing. The proof is similar to that of Theorem 5.

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